Mechanical Engineer, Data Centers - Remote (U.S.)

AECOM · Dallas, TX

Job ID (this collection)	CONSULT-004
Employer	AECOM
Location(s)	Dallas, TX
Career stage	Experienced (6+ years asked)
Location group	DFW area
Majors	Mechanical Engineering
Consulting area	Engineering design & infrastructure consulting
Pay listed	Not listed
Rules stated	No visa sponsorship stated
Recruiting status when checked	Open when checked: the employer page showed the posting on Oct 28, 2026.
Employer page	https://jobs.smartrecruiters.com/AECOM2/744000142816869-mechanical-engineer-data-centers-remote-u-s-
Captured	October 28, 2026

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Full job posting

Company AECOM

Location Dallas, TX, US (Remote)

Job ID J10151918

Posted 2026-08-11

Employment type Full-time

- Experience level
- Associate
- Function
- Engineering
- Company Description

Work with Us. Change the World.

At AECOM, we're delivering a better world. Whether improving your commute, keeping the lights on, providing access to clean water, or transforming skylines, our work helps people and communities thrive. We are the world's trusted infrastructure consulting firm, partnering with clients to solve the world's most complex challenges and build legacies for future generations.

There has never been a better time to be at AECOM. With accelerating infrastructure investment worldwide, our services are in great demand. We invite you to bring your bold ideas and big dreams and become part of a global team of over 50,000 planners, designers, engineers, scientists, digital innovators, program and construction managers and other professionals delivering projects that create a positive and tangible impact around the world.

We're one global team driven by our common purpose to deliver a better world. Join us.

Job Description

AECOM is seeking to hire a highly skilled Mechanical Engineer for the Data Center and Mission Critical team for our east region. The ideal candidate for this position should be familiar with the tech sector, mission critical and data center industry, understanding the requirements of scale, speed, and consistency in data centers.

This position offers the flexibility of a remote work schedule and can be based from a variety of locations in the US where AECOM has an office.

Although the position is primarily remote periodic travel may be required for client meetings, key project milestones, onsite collaboration sessions, or other business-critical activities. Travel is expected to be occasional and scheduled in advance

AECOM is a leader in the global data center consulting industry with robust teams in the Americas, Europe, Africa, Asia, and Australia. In this role, you will have the opportunity to work with our teams around the world on high-profile, fast-paced data center projects for global clients. This is a hands-on, client-facing role with an opportunity for upward career mobility.

Responsibilities, required experience, and qualifications are summarized below:

Responsibilities:

- Data Center Heat Rejection and HVAC System Design: Developing and designing Heat Rejection and HVAC systems specifically tailored for data centers, considering factors such as type of cooling requirement as air cooled or direct fluid cooled, associated cooling loads, future flexibility of the system to accommodate variable heat distribution scenarios, phased construction and operation, airflow management, humidity control, system redundancy providing concurrent maintainability, energy efficiency, and compliance with industry standards and best practices.
- Cooling System Selection: Selecting appropriate air/water side cooling systems for data centers, including Fan wall units, computer room air handlers (CRAHs), chilled water systems, DX systems and CRAC units, in-row cooling, and containment solutions, based on the specific cooling requirements of the data center and the criticality of the equipment.
- Computational Fluid Dynamics (CFD) Analysis: Conducting internal CFD simulations to optimize airflow distribution, transient period analysis, identify hotspots, and ensure efficient cooling throughout the data center. Conducting external CFD on individual buildings and/or campus-wide to simulate effects of different electrical and mechanical components on each other and effects on design day selection criteria capacity of each component. For both types, prepare and/or direct preparation of models and analyse the results to recommend design modifications and optimize the Heat rejection and HVAC system performance. Coordinate the required changes and resulting effects with other trades on project.
- Redundancy and Fault Tolerance: Designing HVAC systems with redundancy and fault-tolerant features to ensure continuous operation of the data center in the event of equipment failure or maintenance activities. This includes the design of backup cooling systems, automatic switchover mechanisms, and monitoring systems.
- Heat Load Calculation and Capacity Planning: Performing heat load calculations for data centers to determine the cooling requirements based on the IT equipment, power density, anticipated growth, and phasing of the construction and operational steps. Conducting capacity planning to ensure the HVAC system can accommodate future expansions and upgrades. Performing HVAC loads for admin and common sections of the facility.
- Energy Efficiency and Sustainability: Implementing energy-efficient HVAC design strategies, such as utilizing economizers, variable speed drives, energy recovery systems, and optimizing airflow management techniques, to minimize energy consumption and support sustainability initiatives.
- Air Quality and Filtration: Ensuring the data center's air quality is maintained at optimal levels by designing appropriate filtration systems and implementing air quality monitoring mechanisms. Addressing concerns related to particulate matter, humidity control, and contaminant removal to protect sensitive IT equipment.
- Coordination with Electrical and Mechanical Systems: Collaborating with electrical engineers and mechanical engineers to integrate the heat rejection and HVAC system seamlessly with the electrical infrastructure, UPS systems, emergency power systems, fire suppression systems, and other critical infrastructure components.

Qualifications

Minimum Qualifications:

- Bachelor's Degree with 6+ years of relevant experience or demonstrated equivalency of experience and or education.
- Experience in data center mechanical and electrical equipment installation and operation.
- P. E. registration is required

Preferred Qualifications:

- A bachelor's degree in mechanical engineering, HVAC engineering, or a related field from an accredited university is typically required. Preference would be given to candidates with a master's degree or advanced coursework in HVAC systems or energy management.
- Knowledge of data center TER Standard Topology as per Uptime Institute, TIA 942, ANSI/BICSI 002
- Professional certifications related to HVAC engineering, such as Certified Energy Manager (CEM), Building Energy Modeling Professional (BEMP), or Leadership in Energy and Environmental Design (LEED) accreditation, would be advantageous.

Additional Information

- Relocation is not available for this position
- Sponsorship is not available for this position
- THIS POSTING IS FOR A CURRENT VACANCY, AND AECOM INTENDS TO FILL THIS POSITION BY NOVEMBER 9, 2026.

At AECOM, we are committed to maintaining a secure and trustworthy recruitment process and take any fraudulent hiring activity seriously. To support this commitment, all newly hired employees are required to attend an in-person Day 1 onboarding at an AECOM office location as a condition of employment.

About AECOM

AECOM is proud to offer comprehensive benefits to meet the diverse needs of our employees. Depending on your employment status, AECOM benefits may include medical, dental, vision, life, AD&D, disability benefits, paid time off, leaves of absences, voluntary benefits, perks, flexible work options, well-being resources, employee assistance program, business travel insurance, service recognition awards, retirement savings plan, and employee stock purchase plan.

AECOM is the global infrastructure leader, committed to delivering a better world. As a trusted professional services firm powered by deep technical abilities, we solve our clients' complex challenges in water, environment, energy, transportation and buildings. Our teams partner with public- and private-sector clients to create innovative, sustainable and resilient solutions throughout the project lifecycle – from advisory, planning, design and engineering to program and construction management. AECOM is a Fortune 500 firm that had revenue of $16.1 billion in fiscal year 2025. Learn more at aecom.com.

What makes AECOM a great place to work

You will be part of a global team that champions your growth and career ambitions. Work on groundbreaking projects - both in your local community and on a global scale - that are transforming our industry and shaping the future. With cutting-edge technology and a network of experts, you'll have the resources to make a real impact. Our award-winning training and development programs are designed to expand your technical expertise and leadership skills, helping you build the career you've always envisioned. Here, you'll find a welcoming workplace built on respect, collaboration and community—where you have the freedom to grow in a world of opportunity.

As an Equal Opportunity Employer, we believe in your potential and are here to help you achieve it. All your information will be kept confidential according to EEO guidelines.